PRIMARY TEACHING NOTES • SCIENCE

INDUCTION PROGRAMME • FOR PRIMARY SCHOOL TEACHERS

Teaching Science in Primary Classes

Learning by looking closely, asking questions and testing ideas - practical notes for Classes 1 to 5 in a government primary school classroom.

A primary child does not learn science from a definition on the board. She learns it when she watches a seed split open, feels which stone is heavier, or discovers that a shadow moves during the day. This booklet turns that simple truth into lessons you can teach tomorrow, with materials you already have.

Observe first

Start every topic with something real to see, touch or smell.

Ask, then test

Let children predict, try, and change their minds.

Use what is near

Schoolyard, kitchen and bazaar materials are enough.

> **Who this is for:** Primary School Teachers (PSTs) teaching General Science / General Knowledge in Classes 1-5, especially in large, mixed-ability and low-resource classrooms. Keep it with your lesson diary.

Contents: How children learn science • Inquiry and the 5E flow • Methods that work • Lesson planning • Concepts by class • Low-cost experiments • Safety and questioning • Assessment and inclusion • Activity bank and checklist

FOUNDATIONS

Children learn science with their hands and senses first

Before a child can understand an explanation, she needs an experience to attach it to. Primary science teaching therefore moves in one direction: from the real thing, to talk about it, to words and drawings that record it - never the other way round.

What this means in the classroom

- **Concrete before abstract.** Show a real leaf before the word "vein". Let children hold ice before teaching "melting".
- **Curiosity is the starting point.** A child's "why?" is not a disturbance. It is the lesson beginning. Collect questions on a corner of the board and return to them.
- **Ideas grow slowly.** A Class 2 child who thinks "heavy things always sink" is not wrong in a careless way - she is using what she has seen. New experience, not correction alone, changes the idea.
- **Talk builds understanding.** Children think aloud in pairs in their home language first, then learn the science word in Urdu and English.

Common misconceptions - and how to handle them

Child says	What is behind it	Teacher response
"The sun moves across the sky."	What the eyes see every day.	Track a shadow hourly. Let the movement itself raise the question.
"Plants eat soil."	Roots go into soil; food goes into mouths.	Grow a seed on wet cotton with no soil at all.
"Big things sink, small things float."	Size and sinking often appear together.	Test a big piece of wood against a small coin in water.
"We breathe only through the nose."	Breathing is mostly invisible.	Hand on chest, count breaths after running in place.

> **Golden rule:** Never laugh at a wrong idea and never erase it too fast. Write it down, test it together, and let the child herself say, "Now I think differently."

METHOD • INQUIRY

Ask, predict, test, explain - the inquiry cycle

Inquiry means children do what scientists do, at their own level: they notice something, ask a question, guess what will happen, test it fairly, and explain what they found. You do not need a laboratory for this. You need a good question and ten unhurried minutes.

The 5E flow for a 35-40 minute period

Stage	Time	What happens - example: "What do seeds need to grow?"
1. Engage	5 min	Show two plants - one healthy, one dried. Ask: "What happened to this one?"
2. Explore	12 min	In groups, children set up seeds: one cup with water, one without; one in light, one in a box.
3. Explain	8 min	Groups report. You introduce the words *germinate* and *moisture* after they have seen the thing.
4. Elaborate	8 min	Apply it: "Why do farmers wait for rain - or water the field - before sowing?"
5. Evaluate	5 min	Exit question: "Your friend keeps seeds in a dry tin. Will they grow? Why?"

The science skills you are building

Process skills

- Observing carefully (all senses, safely)
- Sorting and classifying
- Measuring with hands, string, cups - then with scale and ruler
- Predicting before testing
- Recording with drawings and simple tables

Fair testing

From Class 3 onwards, teach one sentence until it becomes a habit:

"Change one thing, keep everything else the same."

Two cups, same seeds, same soil, same place - only the water is different. That is a fair test.

Keep inquiry manageable: In a class of 40-60, do not give every group different materials. One shared test at the front, with children predicting and recording in their copies, is still inquiry - if their minds are testing, not just their hands.

METHODS

Five methods, each for the job it does best

Method	Best for	Primary example
Demonstration	Anything unsafe, costly or too quick / slow for every child to do	Teacher shows a magnet picking up pins through paper; children predict each time.
Hands-on activity	Skills and concepts children must feel to understand	Sink-or-float testing with a bucket of water and 10 everyday objects.
Observation walk	Living things, weather, shadows, school environment	A 10-minute yard walk: find 5 living and 5 non-living things; draw one.
Talk and storytelling	Topics too big, far or past to observe directly	The journey of a drop of water, told as a story, then drawn as a cycle.
Picture and model work	Inside-the-body topics and invisible processes	A paper model of day and night: one ball, one torch, one dark corner.

Planning a science lesson in 6 steps

1. **One outcome:** "By the end, children can sort objects into floating and sinking and say one reason." One outcome per period is enough.
2. **One real starting point:** an object, a picture from life, or a short problem - not a paragraph read aloud.
3. **Prediction first:** every child commits to a guess, in words or by a show of hands, before any test.
4. **Activity in pairs or groups of 4-5,** with one clear job per child: holder, tester, recorder, reporter, materials keeper.
5. **Record simply:** a drawing, a two-column table, or three sentences with the science words underlined.
6. **Close the loop:** return to the prediction. "Who changed their mind? What changed it?" That moment is the learning.

> **For large classes:** Prepare one materials bag per group the day before, demonstrate the steps once at the front, and appoint materials keepers. The first activity of the year will be slow. The fifth will run itself.

CURRICULUM MAP

Concepts grow step by step from Class 1 to Class 5

Teach the stage the child is at, not the stage the book wishes for. If a Class 4 child cannot yet sort materials confidently, ten minutes of sorting will do more than a rushed new chapter.

Stage	Core concepts	What good teaching looks like
Classes 1-2 Ages 5-7	My body and senses; living and non-living; animals and plants around us; day and night; keeping clean and healthy; sorting by colour, shape, size and material	Mostly oral and practical. Drawing instead of writing. Short activities, frequent movement, nature walks in the schoolyard.
Class 3 Age 8-9	Parts of plants and their jobs; animal homes and food; solids, liquids and gases in daily life; sun, shadow and simple weather; safe and unsafe things at home	First fair tests, first simple tables. Children begin recording results and giving reasons: "because I saw..."
Classes 4-5 Ages 9-11	Life cycles; food chains; states of matter and changes (melting, evaporation); simple machines; light and sound; magnets; earth, moon and seasons; environment and cleanliness of water and air	Longer investigations over several days, measuring with standard units, presenting group findings to the class.

Spiral teaching: revisit, do not repeat

The same big idea returns every year at a deeper level. Plants in Class 1 means naming leaf and flower. In Class 3 it means what each part does. In Class 5 it means how a plant makes food and why green belts matter for a village. When you know where a topic is going, today's lesson has a purpose beyond today's page.

Connect to daily life

- Evaporation: wet clothes drying on a line
- Insulation: a clay pot keeping water cool
- Levers: a seesaw, a wheelbarrow, opening a bottle
- Germs: handwashing before the midday meal

Connect across subjects

- Maths: measuring, counting, simple graphs
- Urdu / English: new science words, describing
- Art: careful observational drawing
- Islamiat / values: care for animals, plants and water

PRACTICAL WORK

Real experiments with low-cost and no-cost materials

You do not need a science kit to teach science well. You need a small, permanent corner box that you build up over one term.

The primary science box - build it once

Plastic bottles and cups, old jars, candles (teacher use only), magnets, a torch, string, rubber bands, balloons, straws, spoons, coins, stones, seeds (moong, gram, wheat), cotton, soil in a bucket, a hand lens if available, old newspapers, and a bucket for water work. Ask children to bring clean, safe items from home - never glass, blades or chemicals.

Six dependable experiments

Experiment	Materials	Concept it teaches
Growing seeds on cotton	Cup, cotton, moong seeds, water	Germination; seeds need water and air, not soil, to start
Sink or float?	Bucket of water, 10 safe objects	Floating depends on material and trapped air, not size alone
Shadow clock	Stick, chalk, schoolyard sun	The sun appears to move; shadows change through the day
Which keeps ice longer?	Ice, cloth, plastic, newspaper	Insulation - some materials slow melting
Magnet hunt	One magnet per group	Some metals are magnetic; force works through paper and wood
Evaporation race	Two wet cloths - sun and shade	Heat and air speed up evaporation

> **Safety - non-negotiable rules:** No tasting in science, ever. No smelling unknown substances. Teacher alone handles fire, boiling water and sharp objects. Wash hands after soil, seeds and outdoor work. No batteries opened, no mains electricity experiments, no chemicals from home. Stop any activity the moment it becomes throwing, splashing or crowding around water.

Outdoor rule: agree the boundary and the signal to return before you leave the classroom. One whistle or raised hand, practised once, saves ten minutes of calling names.

TALK, ASSESSMENT, INCLUSION

Good questions, honest assessment, every child included

Questioning that makes children think

Ask more of these

- "What do you notice?"
- "What do you think will happen? Why?"
- "How could we test that?"
- "What changed? What stayed the same?"
- "Do you still think what you thought before?"

Ask less of these

- Questions only one child can answer from memory
- "Is this clear?" - everyone says yes
- Rapid-fire questions with no wait time

Wait 5 full seconds after a good question. Silence is thinking time.

Language support

Introduce every science word in three steps: show the real thing, say the word in the child's home language and Urdu, then write the English word with a small drawing beside it. Keep a growing science word wall for use during recording. Never punish a correct science idea for weak English; assess the idea first.

Assessing science without fear

Tool	How to use it
Observation checklist	Watch 5-6 children per lesson: observing? predicting? recording? One line each; the whole class is covered in a week.
Draw and label	"Draw what happened and label two things." Reveals understanding that writing hides, especially in Classes 1-3.
Exit question	One question at the door. Sort into "got it / partly / not yet" and reteach the "not yet" group first next day.
Show and tell	A group presents its finding in 1 minute. Assess the explanation, not confidence.

Including every learner

- **Slower learners:** fewer objects to sort, a partner recorder, success in the hands-on step before writing.
- **Fast learners:** a harder question, not more of the same - "Would it still float in salty water?"
- **Shy children:** pair-talk first, so an answer is rehearsed before it is public. **Children with disabilities:** a defined job in every group and materials within reach - participation, not watching.

READY TO USE

Activity bank: ten lessons that need almost nothing

1. **Sense walk (Cl. 1-2):** In the yard, find something rough, smooth, smelly, noisy and green. Draw each one.
2. **Living or not? (Cl. 1-2):** Picture cards or real objects sorted into two hoops or chalk circles, with reasons.
3. **Leaf collection (Cl. 2-3):** Collect fallen leaves only. Sort by size, then by edge, then draw veins.
4. **Sound makers (Cl. 3):** Rice in a closed tin, rubber band on a box, water in bottles - what vibrates?
5. **Clean water test (Cl. 3-4):** Filter muddy water through cloth and cotton. Discuss: filtered is not the same as safe to drink.
6. **Paper helicopter (Cl. 4):** Drop paper spinners from the same height. Change one thing - size, or a paper clip - and time the fall.
7. **Day and night model (Cl. 4):** Torch and ball in a dark corner. Mark home with a dot; turn slowly.
8. **Food chain cards (Cl. 4-5):** Grass, grasshopper, hen, human - arrange, then remove one card. What happens?
9. **Dissolving race (Cl. 5):** Salt in cold water vs warm water, same stirring. Record times in a table.
10. **Schoolyard audit (Cl. 5):** Count litter, leaking taps and bare ground. Present one improvement the class can actually do.

Teacher checklist - before, during, after

[ ] One clear outcome written in my diary, in child-friendly words

[ ] Materials tested by me first - the experiment works before the class tries it

[ ] Prediction asked before any demonstration or activity begins

[ ] Every child has a job or a recording task; no one only watches

[ ] Safety rule for this activity stated out loud

[ ] Science words introduced after the experience, with a drawing

[ ] Exit check done - I know who needs reteaching tomorrow

[ ] Materials collected, counted and back in the science box

> **One last word:** You are not behind because you have no laboratory. A teacher who asks a good question, lets children test it with a bucket, a seed and a stick, and listens carefully to what they say, is teaching real science. Start with one activity this week.

Companion notes in this series cover classroom arrangement, child psychology, pedagogy, teaching mathematics, and ICT in the primary school - each as a separate booklet.